

SE LAB1 SUBMISSION

NAME : DIRAN SOMANNA

SRN : PES1UG24AM364

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PES University - Dept. of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #06 | Campus & Academic Operations

Alumni Mentorship & Mock Interview Platform

1. Problem Context & Overview Connecting students with alumni working in specialized domains requires an intelligent matching mechanism, availability calendar booking, and structured post-interview scorecard recording.

Target Stakeholders / Actors: Student Mentee, Alumni Mentor

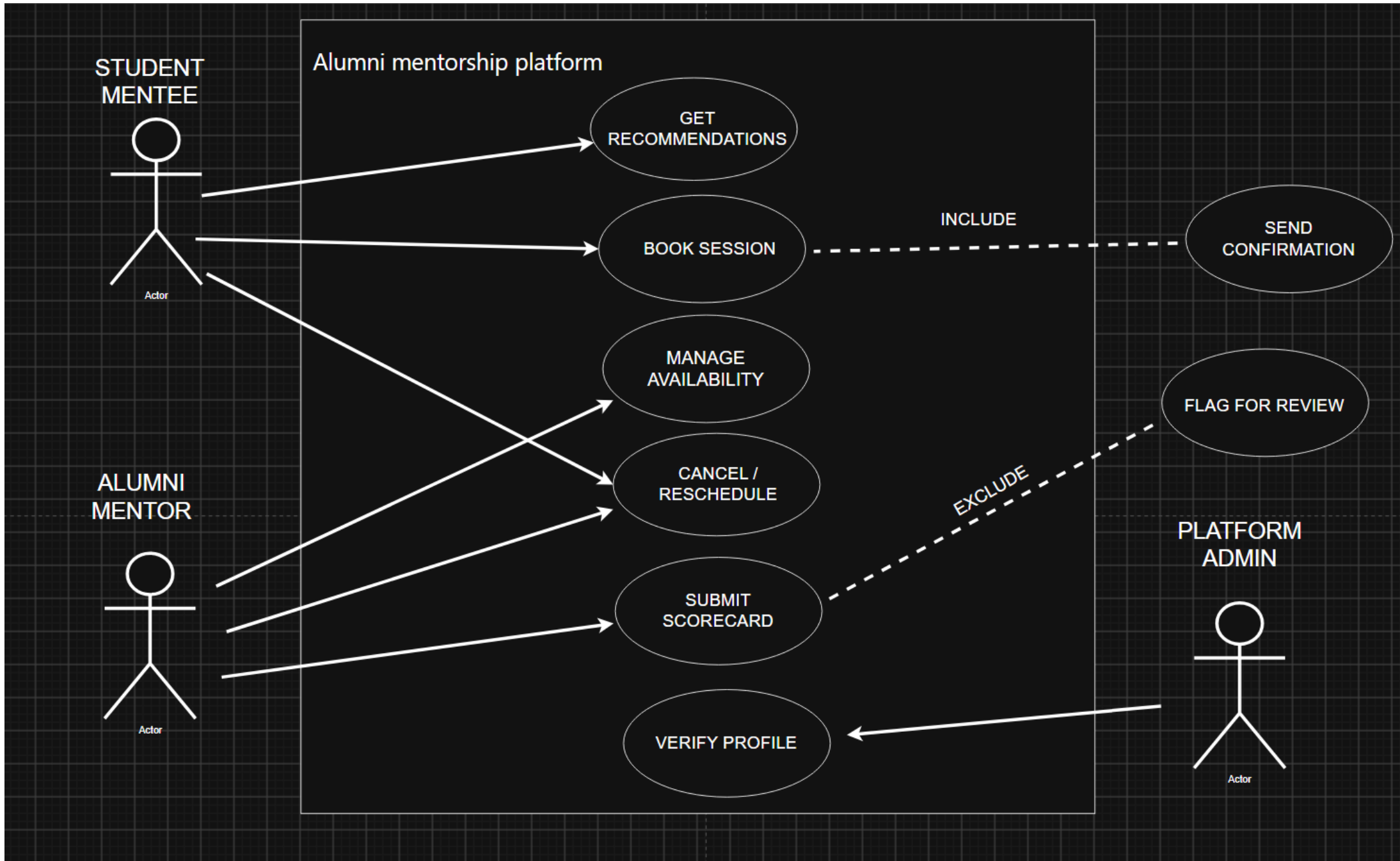
• Complete Requirements Table:

>> Exactly 5 Functional Requirements (FR-001 to FR-005)

>> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing) and facilitate one-click session booking.	High	Pass: Mentorship session is added to both calendars with meeting link; Fail: Double-booking of mentor slots.	Core matching functionality drives platform adoption.
FR-002	Functional	The system shall allow an alumni mentor to set and update their weekly availability slots on a calendar.	High	Pass: Updated slots reflect immediately in student search results; Fail: Stale/removed slots still appear as bookable.	Availability accuracy prevents booking conflicts.
FR-003	Functional	The system shall allow a student mentee to cancel or reschedule a booked session at least 12 hours in advance.	Medium	Pass: Cancelled/rescheduled slot is released back to the mentor's calendar within 5 minutes; Fail: Slot remains locked after cancellation.	Enables flexibility without wasting mentor time.
FR-004	Functional	The system shall allow a mentor to submit a structured post-interview scorecard (ratings + comments) after each mock interview session.	High	Pass: Scorecard is saved and visible to the student within 1 minute of submission; Fail: Scorecard fails to save or is not linked to the correct session.	Structured feedback is the core value delivered to students.
FR-005	Functional	The system shall send an automated reminder notification (email/push) to both mentor and mentee 24 hours before a scheduled session.	Medium	Pass: Notification is logged as sent 24±1 hours before session start; Fail: No notification sent or sent at wrong time.	Reduces no-shows for scheduled sessions.
NFR-001	Nonfunctional	All user profile data and feedback scorecards shall adhere to privacy guidelines and support soft-deletion within 24 hours of request.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given — protects sensitive personal/feedback data.
NFR-002	Nonfunctional	The mentor recommendation engine shall return a ranked list of matching mentors within 2 seconds for 95% of requests under normal load.	High	Pass: Load testing logs show 95th percentile response time ≤ 2s across 500 concurrent requests; Fail: Response time exceeds 2s for >5% of requests.	Slow matching would hurt usability and adoption.

- UML Use-Case Diagram: >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.



- **Use-Case Flow Specification:**

>> **1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario**

>> **One Alternate Flow.**

Use-Case Flow Specification

Use Case: Book Mentorship Session

Actors: Student Mentee, Alumni Mentor

Preconditions

1. Student is logged in with a verified profile.
2. At least one alumni mentor matching the student's selected domain has open availability slots.

Postconditions

1. The mentorship session is added to both the student's and mentor's calendars with a meeting link.
2. A confirmation notification is sent to both parties.

Main Success Scenario

1. Student selects "Find a Mentor" and views a list of domain-matched mentor recommendations.
2. Student selects a mentor and views their available time slots.
3. Student selects a slot and confirms the booking.
4. System validates that the selected slot is still available.
5. System books the slot and creates calendar entries for both the mentor and the student.
6. System generates a meeting link for the session.
7. System sends a confirmation notification to both the mentor and the student.
8. Use case ends successfully.

Alternate Flow

- 4a. Slot No Longer Available (Double-Booking)

- 4a1. System displays a "Slot no longer available" message to the student.
- 4a2. System refreshes the mentor's list of available slots.
- 4a3. Student selects a different slot or a different mentor.
- 4a4. Use case resumes at step 3.